STUDENT ACTIVITY

Evolution and Adaptation — Activity

Activity: Regression Model Selection and Adaptation

Q: Read the prediction scenario below and describe how three different regression models would attempt to process the variables to predict the outcome:

- One approach using Simple Linear Regression.
- One approach using Multiple Linear Regression.
- One approach using Polynomial Regression.

Next, based on the provided text, describe why a data scientist might apply Ridge or Lasso Regression to a complex dataset in three sentences. Ensure you explicitly mention overfitting, regularization, and feature selection.

The Scenario: A real estate analytics company is building a machine learning model to predict house prices accurately. Initially, the team attempts to predict the price based solely on the area of the house, assuming a basic straight-line relationship between the two variables. However, they soon realize that the price trend actually follows a curved, nonlinear relationship as properties get exceptionally large. To further maximize prediction accuracy, the team abandons the single-variable approach and decides to consider several independent factors simultaneously, including the area, number of rooms, location, and age of the house.

Activity: Regression Model Selection and Adaptation

Regression Model Selection

Simple Linear Regression

Regression Model	How it Processes the Variables	Best Use in the Scenario
Simple Linear Regression	Uses **one independent variable** (house area) to predict **one dependent variable** (house price). It assumes a **straight-line (linear) relationship** between the variables.	Suitable when predicting house prices **only from the area** and the relationship is approximately linear.

Multiple Linear Regression

Regression Model	How it Processes the Variables	Best Use in the Scenario
Multiple Linear Regression	Uses **multiple independent variables**, such as **area, number of rooms, location, and age of the house**, to predict the house price. It combines the influence of all variables to improve prediction accuracy.	Suitable when **several factors** affect house prices and a more accurate prediction is required.

Polynomial Regression

Regression Model	How it Processes the Variables	Best Use in the Scenario
Polynomial Regression	Extends linear regression by adding **polynomial terms** (such as x2x^2x2, x3x^3x3) to model **curved or nonlinear relationships** between the independent variable and the target variable.	Suitable when house prices increase in a **curved (nonlinear)** pattern as the house area becomes very large.

Why Apply Ridge or Lasso Regression?

A data scientist applies **Ridge Regression** or **Lasso Regression** to reduce **overfitting** when working with complex datasets that contain many features. Both methods use **regularization**, which adds a penalty to the regression model to reduce the impact of unnecessary or overly large coefficients and improve generalization. **Lasso Regression** also performs **feature selection** by shrinking some coefficients to zero, automatically removing less important features from the model.

Regression Models

Regression Model	How It Processes the Variables	Application in the Scenario
Simple Linear Regression	Uses **one independent variable (house area)** to predict house price with a **linear relationship**.	Predicts price based only on **house area**.
Multiple Linear Regression	Uses **multiple independent variables** such as **area, number of rooms, location, and age** to predict house price.	Produces **more accurate predictions** by considering several influencing factors.

Regression Model	How It Processes the Variables	Application in the Scenario
Polynomial Regression	Models a **nonlinear (curved) relationship** by adding polynomial terms to the regression equation.	Captures the **curved price trend** for very large houses.

Ridge and Lasso Regression

- **Ridge Regression** and **Lasso Regression** help prevent **overfitting** in complex datasets.
- They use **regularization** to reduce the effect of large model coefficients and improve prediction performance.
- **Lasso Regression** also performs **feature selection** by setting unimportant feature coefficients to zero, creating a simpler and more efficient model